

SUDHEESH SREENILAYAM

Data Systems & AI Architecture

Architecting End-to-End AI Systems & High-Density Data Pipelines

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SUMMARY

AI-Native Systems Architect and Data Engineer with 8+ years of progressive operational experience. **I bridge the gap between deep technical implementation and high-level business strategy.** I engineer automated ETL pipelines, deploy agentic AI workflows, and design decision engines that reduced operational waste by **\$900k annually**. I translate complex data environments into scalable, enterprise-level solutions.

SKILLS

Data Architecture & Databases: SQL (PostgreSQL, BigQuery, Snowflake), dbt, Git LFS, Databricks, Supabase, Hadoop, Hive.

AI Ecosystem & Automation: Agentic AI (LangChain, Claude, OpenClaw, Antigravity), Python (Pandas, Scikit-learn), n8n, Zapier.

Machine Learning & Analytics: SVM, NLP (TF-IDF), SMOTE, Regression Trees, Random Forest, Power BI, Tableau.

PROFESSIONAL EXPERIENCE

Senior Associate Manager, Operations Analytics

Sep 2023 – Aug 2024

Sutherland Global Services (Amazon Logistics)

- AI Solutions Architecture:** Architected the "Driver Self-Service" decision engine using SQL and Power Query, eliminating a 66% defect rate across 500k+ tickets and driving **\$900k in annual savings**.
- Reporting Infrastructure:** Deployed automated Power BI dashboards across the global delivery center, eliminating manual data latency.
- System Ownership:** Translated stakeholder requirements into hard technical specifications, ensuring successful UAT across NA, UK, and CA regions.

Associate Manager, Process Improvement

Dec 2021 – Aug 2023

- Algorithm Design:** Engineered a predictive staffing allocation algorithm in Excel to balance shift coverage for 90+ FTEs, reducing team overtime costs by 15%.
- Statistical Analysis:** Led a cross-functional task force utilizing Minitab to perform hypothesis testing on service bottlenecks, driving a 2.73% operational throughput gain.

Lead, Service Delivery Management

Dec 2019 – Nov 2021

- Data Pipelines:** Automated manual performance tracking and streamlined data collection pipelines, saving front-line leadership 5+ hours of labor weekly.
- A/B Testing:** Designed and executed rigorous A/B tests on customer service scripts, driving a **16% improvement in FCR**.

Early Career (Promoted from Associate to Senior SME)

Jun 2016 – Nov 2019

- Data Monitoring:** Achieved 100% SLA compliance by developing real-time monitoring reports via Excel Power Query.

STRATEGIC PROJECTS & INNOVATION

Chicago 311 Accountability Dashboard

Python, ETL, React, D3.js, GitHub Actions

- Automated the nightly processing of **1M+ records** from the Chicago Open Data Portal to investigate urban service inequities by engineering a cost-efficient "Static-Site Data Orchestration" system.

Fake Job Posting Detection (Machine Learning)

Python (NLP, SMOTE), SVM, Scikit-learn

- Achieved a top **F1-Score of 0.84** in fraud detection by managing the end-to-end development of an ML pipeline and addressing a 95% class imbalance using SMOTE and SVM classification.

Distributed Systems & MapReduce Implementation

AWS EMR, Hadoop, Hive, HBase

- Optimized high-volume trade price aggregation and query performance by designing distributed computing tasks leveraging HDFS Architecture.

EDUCATION

Master of Science in Business Analytics

May 2026

Roosevelt University, Chicago, IL

GPA: **3.97/4.0**

B.Tech in Electronics & Communication Engineering

CUSAT, India

CERTIFICATIONS

- Lean Six Sigma Green Belt (Active)
- Google Data Analytics (In Progress)